Portfolio Project Brief

Integrating Machine Learning and AI to Drive Business Value
Advanced Software Engineering - Module 4

How to use this brief:

Read this document in full and bring any questions you have on the assessment to either a workshop or the next 1:1 with your instructor. **Advance in your project journey by completing milestones**, which are progress checks on tasks that build to your overall output. Milestone submission and reviews will be part of your workshop and independent work. **Self-assess your progress** against the assessment criteria table while completing your project.

Module Title	Integrating Machine Learning and AI to Drive Business Value		
Format	Problem-based report	**Deadline**	On MyMultiverse
Length	4000 words	**Feedback Due**	Within 20 working days of submission date
Intended Audience	Senior Leaders in your organization (but delivered to your instructor)	**Weighting**	100%
Project Title	Machine Learning Solution(s) Report		
Project Learning Outcomes	1. Demonstrate a critical understanding of machine learning and AI principles, integrating these concepts into organizational digital technology solutions' lifecycle and fulfilling organizational standards and tools (K6, K7, K13). 2. Develop software application(s) for machine learning and AI, emphasizing coding, testing and debugging in compliance with design requirements (K24, S4). 3. Implement machine learning models in diverse scenarios by applying appropriate data analysis techniques and initiating a data product design that fits the digital technology solution needs (S10, S11). 4. Work reliably and capably within teams on machine learning and AI projects, upholding legal, ethical and social standards, while sharing best practices within the organization and the wider community (S7, K19, S15, B2, B6).		
Project Milestones	**Milestone 1: Value Proposition of Your Learnings (Weeks 1-3)** In the first three weeks, you will build a foundational understanding of ML and AI		

	principles. Submit a report including an executive summary, introduction, and comprehensive explanation of ML and AI concepts, focusing on their integration into digital solutions and compliance with organizational standards. **Milestone 2: Value Creation through Application (Weeks 4-7)** From weeks 4 to 7, you will apply these concepts to develop a practical ML solution(s). Submit a progress report and a working prototype, detailing the development process, model selection, data product design, and considerations for ethics and regulatory compliance. **Milestone 3: Impact Assessment (Weeks 8-9)** In the final weeks, you will assess the impact of your work within your organization. Submit the final Change Management Guidelines, including an impact assessment, ethical considerations, collaboration plan, and recommendations for stakeholders, along with supplementary materials in the appendices.
Feedback	Written feedback and a grade will be given within 20 working days after the deadline. Opportunity for further discussion will be available during individual coaching sessions.

Why is this project important to me and my organization?

This project is important for both personal and professional development as it will provide a comprehensive understanding of machine learning and AI principles and their practical application in software development. The project will also emphasize ethical considerations, promoting responsible practice in the use of these technologies.

For the organization, the project could have a significant impact by providing a practical framework for integrating ML into organizational change management, promoting innovation while maintaining ethical integrity and compliance with regulatory requirements. This could lead to improved processes, operations, and outputs, enhancing the organization's overall performance and competitiveness. Additionally, the developed guidelines could be shared within the organization and the wider community, promoting best practices and responsible use of ML and AI technologies.

What do I need to do for this project?

For this project, you will develop comprehensive Change Management Guidelines, incorporating machine learning (ML) principles and artificial intelligence (AI) technologies. The guidelines will be derived from a real business problem or opportunity for change within the organization, leveraging ML to enhance processes, operations, or outputs. Your task is to create a structured approach that encompasses best practices for implementing ML solutions, ensuring these solutions are not only effective but also ethically sound and compliant with relevant standards.

To successfully complete this project, you will need to demonstrate a critical understanding of machine learning and AI principles, integrating these concepts into digital technology solutions' lifecycle and fulfilling organizational standards and tools. You will also need to develop

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software application(s) for machine learning and AI, emphasizing coding, testing and debugging in compliance with design requirements. Additionally, you will implement machine learning models in diverse scenarios by applying appropriate data analysis techniques and initiating a data product design that fits the digital technology solution needs.

Finally, you will work reliably and capably within teams on machine learning and AI projects, upholding legal, ethical and social standards, while sharing best practices within the organization and the wider community. The developed guidelines will offer a practical framework for integrating ML into organizational change management, promoting both innovation and responsible practice. This framework will support organizations in harnessing the power of ML and AI while maintaining ethical integrity and compliance with regulatory requirements.

Tools/Software

The software you should use for this assignment should reflect those utilized in the workshops, so you should demonstrate your ability to use Python and Jupyter Notebook (or equivalent) to develop the machine learning model.

Final Project Output

The overall project output is a set of comprehensive Change Management Guidelines that incorporate machine learning (ML) principles and artificial intelligence (AI) technologies to solve a real business problem or opportunity for change within the organization. The guidelines must encompass best practices for implementing ML solutions and should promote innovation while maintaining ethical integrity and compliance with regulatory requirements.

*Please note: The following section guidelines are just suggestions. Please add, amend, or structure to best fit your project.

Section	Content Covered
1. **Executive Summary and Introduction** (~500 words; LO1)	• Provide a high-level **overview of the guidelines,** the business problem addressed, and the importance of integrating ML and AI. • Discuss the **significance of ML and AI** integration in organizational change management. • Outline the **scope** of the report and the main **objectives**. • Highlight the anticipated outcomes and benefits.
2.**Machine Learning and AI principles** (~700 words; L01)	**Key Concepts and Principles** • Explain the fundamental principles of ML and AI. • Discuss the key technologies and methodologies relevant to the business problem. **Integration into Digital Solutions Lifecycle** • Analyze how these principles are incorporated into the digital technology solution lifecycle.

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	• Outline a plan to ensure compliance with organizational standards and tools.
3: **Developing ML and AI software Application(s)** (~1000 words; LO2)	**Application Development** • Briefly introduce the application(s) you are developing and the problem that it solves. • Describe the development process, including coding, testing, and debugging. • Emphasize compliance with design requirements and organizational standards. **Model Selection and Methodologies** • Discuss the criteria for selecting the appropriate ML model(s) for your solution. • Describe the methodologies and techniques used in model development.
4. **Implementation and Data Product Design** (~1000 words: LO3/LO4)	**Data Analysis Techniques** • Research and analyze various ML models and data analysis techniques. • Identify the most suitable models and techniques for the identified business problem. **Designing a Data Product** • Outline the design of a data product tailored to the digital solution needs. • Ensure it aligns with business objectives and user requirements. **Ethical Considerations** • Discuss the ethical implications of ML and AI deployment. • Outline steps taken to mitigate biases and ensure fairness in the model. **Regulatory Compliance** • Ensure compliance with data protection regulations and other legal standards. • Describe measures to safeguard data privacy and security.
5. **Team Collaboration and Communication plan** (~400 words: LO4)	**Collaboration Plan** • Develop a strategy for effective team collaboration on ML and AI projects.

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	• Highlight roles, responsibilities, and communication channels. **Sharing Best Practices** • Detail how best practices will be shared within the organization and the wider community. • Emphasize continuous learning and adaptation.
Recommendations and Conclusion (~400 words)	• **Summarize the key points** discussed in the report. • Reaffirm the strategic importance of ML and AI integration. • **Provide specific recommendations** for senior stakeholders on implementing these guidelines. • Focus on prioritizing projects, resource allocation, and ongoing evaluation.
Appendices (word count N/A)	Include any additional information, such as detailed research findings, technical specifications, or supplementary materials that support the main content of the report.

Need help getting started?

Have a look at these example projects:

1. **Fraud Detection**: Build a machine learning model that analyzes transaction data and customer behavior to identify and flag potentially fraudulent activities in real-time for an e-commerce platform.
2. **Demand Forecasting:** Develop a machine learning model that predicts future product demand across store locations, considering various factors, to optimize inventory management and supply chain efficiency for a retail chain.
3. **Sentiment Analysis:** Create a machine learning model that performs real-time sentiment analysis on social media posts and customer reviews to provide insights on brand perception for a social media management company.
4. **Predictive Maintenance:** Develop a machine learning model that predicts equipment failures based on various data sources to implement proactive maintenance strategies, reducing downtime and optimizing schedules for a manufacturing company.

Project Milestones

Project milestones are completion points for the overall output of your module project. Please note that the project is intended to be a piece of applied work that develops over the course of

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the module, rather than a summative assessment to be completed in a short, intense period of time at the end of module learning.

NOTE: For step-by-step guidance on milestone completion, follow the Milestone activities for the module on your MyMultiverse account.

Check Your Durable Skills

In addition to the assessment criteria, all assessments across your apprenticeship will also be marked on your durable skills. Use this checklist to ensure you are submitting a high quality piece of work.

- **Clear communication.** I've clearly and concisely communicated my findings and demonstrated professional language, spelling and grammar, and adhered to the word count. Please note that any referencing should be included at the end of the slide show, please refer to the style guide on Applied for further guidance.
- **Thinking strategies.** I've engaged in the critical analysis of available facts, evidence, observations, and arguments to form a judgment on what I think is important. Ultimately I've framed my analysis in a way that arrives at a solution.
- **Self-reflection.** I've considered feedback from previous work to inform the current submission. With this, I am engaging with feedback with a growth mindset in order to cultivate an attitude of continuous improvement.

How should I submit this Portfolio Project?

Your submission should be prepared as a single PDF file. Please make sure your work adheres to the MV referencing and style guide (see MyMultiverse). All work is checked for plagiarism using Turnitin.

Ensure that you meet the MV guidelines for academic integrity by considering the following:

- **Referencing:** When discussing someone else's work, clearly acknowledge the work by using in-text citations and a reference list adhering to the Harvard referencing style.
- **Word Count:** Any text exceeding the maximum word limit will not be marked. Text that adds to the word count includes all text except the reference list or appendices. All text in diagrams and in-text citations count towards the overall word limit.
- **Submission time:** All submissions must be made by mid-day on the specified deadline.

Please name your assignment submission file using the following format **<ULN_ModuleNumber_AssessmentNumber>** (e.g. 0012345_Module2_Assessment2).

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How will my assessment be marked?

Carefully review the assessment criteria as you complete milestones and outputs of your project.

Description	F	E	D	C	B	A
Demonstrate a critical understanding of machine learning and AI principles, integrating these concepts into organizational digital technology solutions' lifecycle and fulfilling organizational standards and tools (K6, K7, K13).	**Fails to identify or explain basic ML/AI principles** relevant to the digital solution lifecycle. Shows no understanding of organizational standards or implementation approaches.	**Recognizes basic ML/AI principles** but struggles to relate them to the digital solution lifecycle. Shows limited awareness of organizational standards and implementation needs.	**Identifies basic ML/AI principles** relevant to the digital solution lifecycle. **Outlines a simple plan** to align with organizational standards. Provides minimal rationale for choices.	**Identifies basic principles of machine learning and AI** relevant to the digital technology solution lifecycle. **Outlines a simple plan** to align with organizational standards and tools.	**Analyzes key ML/AI principles** relevant to the digital solution lifecycle. **Develops a detailed implementation plan** aligned with organizational standards. Provides clear rationale for choices and approach.	**Critically evaluates and integrates advanced ML/AI principles** throughout the digital solution lifecycle. **Designs a strategic implementation plan** that enhances organizational standards. Provides in-depth analysis of business impacts, risks, and mitigation strategies.
Develop software application(s) for machine learning and AI, emphasizing coding, testing and debugging in compliance with design requirements (K24, S4).	**Unable to develop a** software prototype for machine learning and AI, and fails to emphasize coding, testing, and debugging in compliance with design requirements.	**Attempts basic development** of a software prototype is evident, but lacks specific alignment with design requirements and organizational standards and guidelines.	**Develops a simple prototype**, describing coding, testing, and debugging in line with design requirements, and ensures compliance with organizational standards and guidelines.	**Develops a working application(s) prototype** for machine learning and AI that are relevant to the design requirements, and provides a plan to ensure compliance with organizational standards and guidelines.	**Designs and implements a robust software application(s)** for machine learning and AI that emphasize coding, testing, and debugging in compliance with design requirements, and ensures compliance with organizational standards and guidelines.	**Creates and delivers an innovative software application(s)** for machine learning and AI that meet design requirements, and demonstrates a deep understanding of how to ensure compliance with organizational standards and guidelines.

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Implement machine learning models in diverse scenarios by applying appropriate data analysis techniques and initiating a data product design that fits the digital technology solution needs (S10, S11).	**Unable to research and analyze** various machine learning models and data analysis techniques, and fails to determine which models and techniques are most appropriate for the solution needs.	**Basic research and analysis** of various machine learning models and data analysis techniques is evident, but lacks specific alignment with the digital technology solution needs.	**Researches and analyzes** various machine learning models and data analysis techniques, determines which models and techniques are most appropriate for the solution needs, and designs a data product that fits the digital technology solution needs.	**Proposes relevant machine learning models** and data analysis techniques that are most appropriate for the digital technology solution needs, and provides a plan to design a data product that fits the solution needs.	**Provides a structured set of machine learning models** and data analysis techniques that are well-aligned with the digital technology solution needs, and includes clear explanations, analysis, and justification.	**Develops a sophisticated set of machine learning models** and data analysis techniques, provides a detailed set of models and techniques that are strategically aligned with the digital technology solution needs, and includes persuasive rationale and clear demonstration of potential business impacts.
Work reliably and capably within teams on machine learning and AI projects, upholding legal, ethical and social standards, while sharing best practices within the organisation and the wider community (S7, K19, S15, B2, B6)..	**Unable to develop a plan** for reliable and capable team collaboration, and fails to maintain legal, ethical, and social standards throughout the project.	**Basic development of a plan** for reliable and capable team collaboration is evident, but lacks specific alignment with legal, ethical, and social standards throughout the project.	**Develops a plan** for reliable and capable team collaboration, maintains legal, ethical, and social standards throughout the project, and shares best practices within the organization and wider community.	**Proposes relevant plan** for reliable and capable team collaboration that is aligned with legal, ethical, and social standards throughout the project, and provides a plan to share best practices within the organization and wider community.	**Provides a structured set of plans** for reliable and capable team collaboration that are well-aligned with legal, ethical, and social standards throughout the project, and includes clear explanations and justification.	**Demonstrates a comprehensive understanding of plan**s for reliable and capable team collaboration that are strategically aligned with legal, ethical, and social standards throughout the project, and provides persuasive rationale and clear demonstration of potential business impacts.

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